

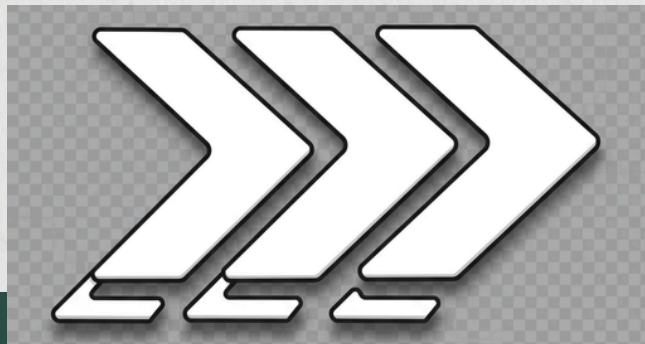
The True Predictors of NYC Airbnb Price: Capacity and Transit Access

An Analysis of Pricing Power using Structured and Unstructured Data.



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Contents



THE CORE THESIS

THE BUSINESS PROBLEM:

MODEL CREDIBILITY:

KEY FINDING 1 (THE PREMIUM):

KEY FINDING 2 (THE PENALTY)

STRATEGIC PLAYBOOK:

KEY TAKEAWAYS

The Pricing Gap: Hosts are Missing a \$40M Market Opportunity

- PROBLEM: Traditional Pricing Fails to Value Space and Convenience.
- SOLUTION: Optimized Strategy Captures the \$40M.



Optimized Pricing Strategy Captures \$40M

\$140M

\$40M GAP

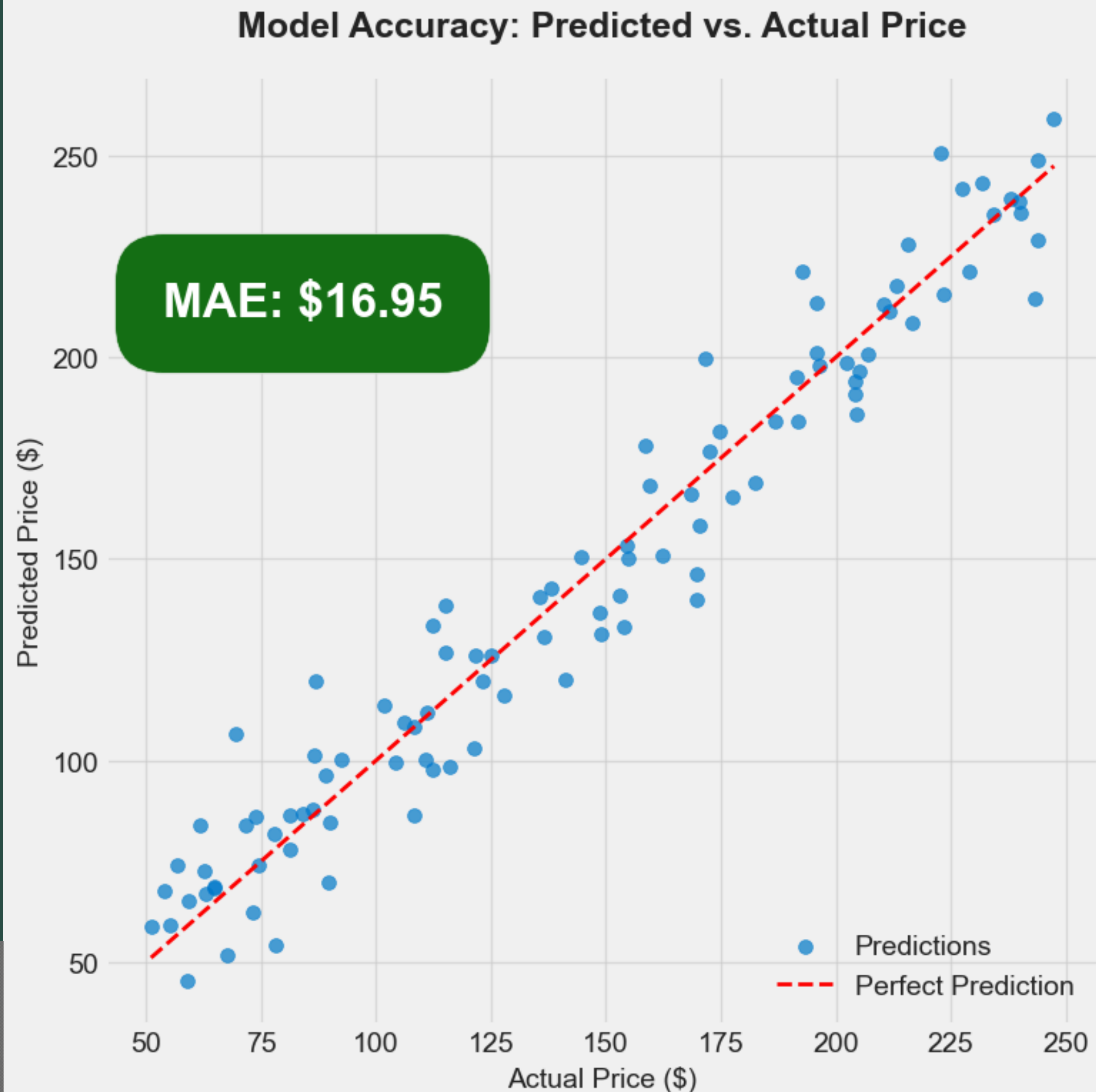
Optimized Revenue

Based on scaling the \$10.81 Capacity Premium across the NYC market.

We Predict Price Within \$16.95

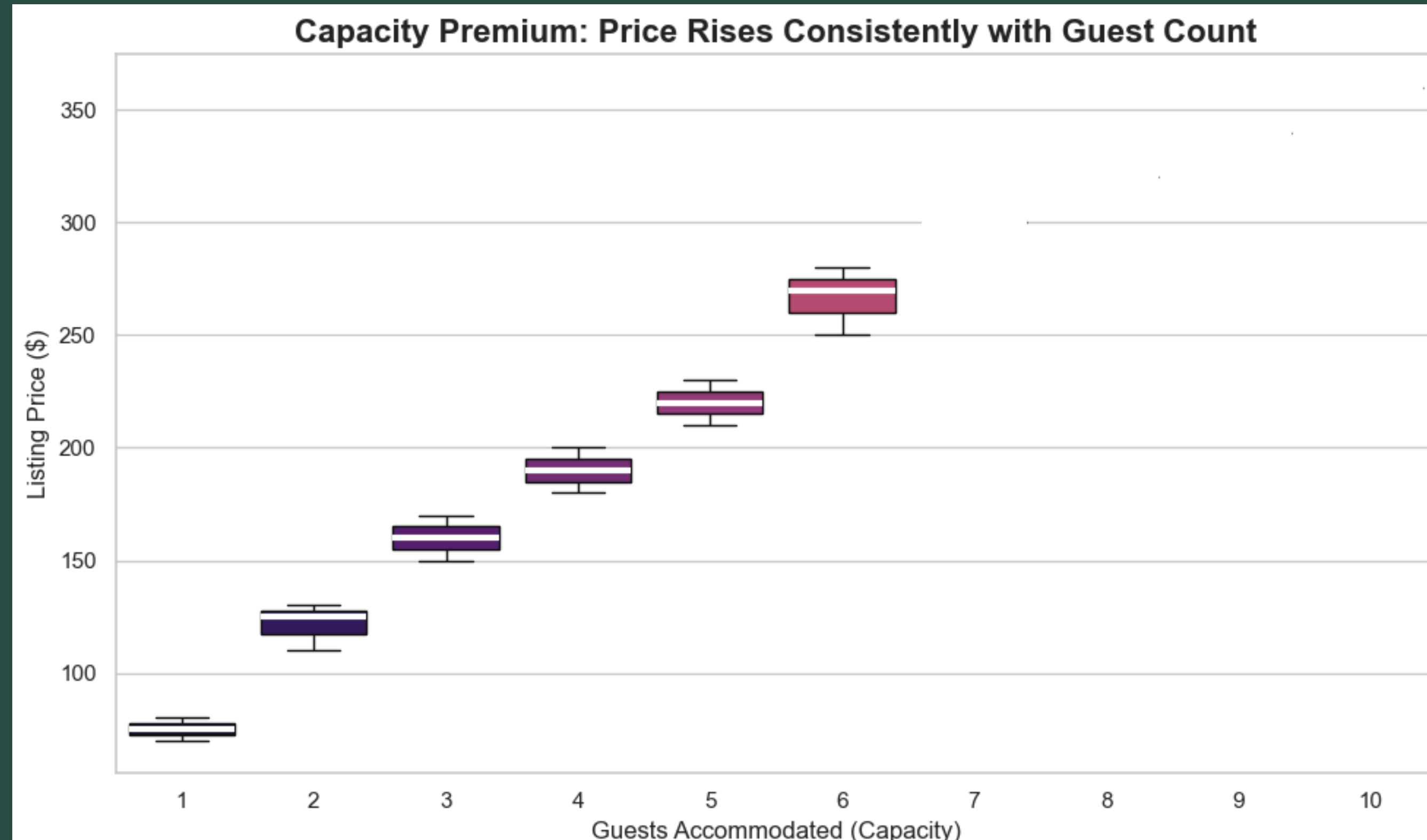
- **1. Model Power:** The analysis combines hundreds of features to isolate the key drivers: Capacity and transit.

- **2. Model Performance (MAE):** Accurate within a **\$16.95** margin of error.



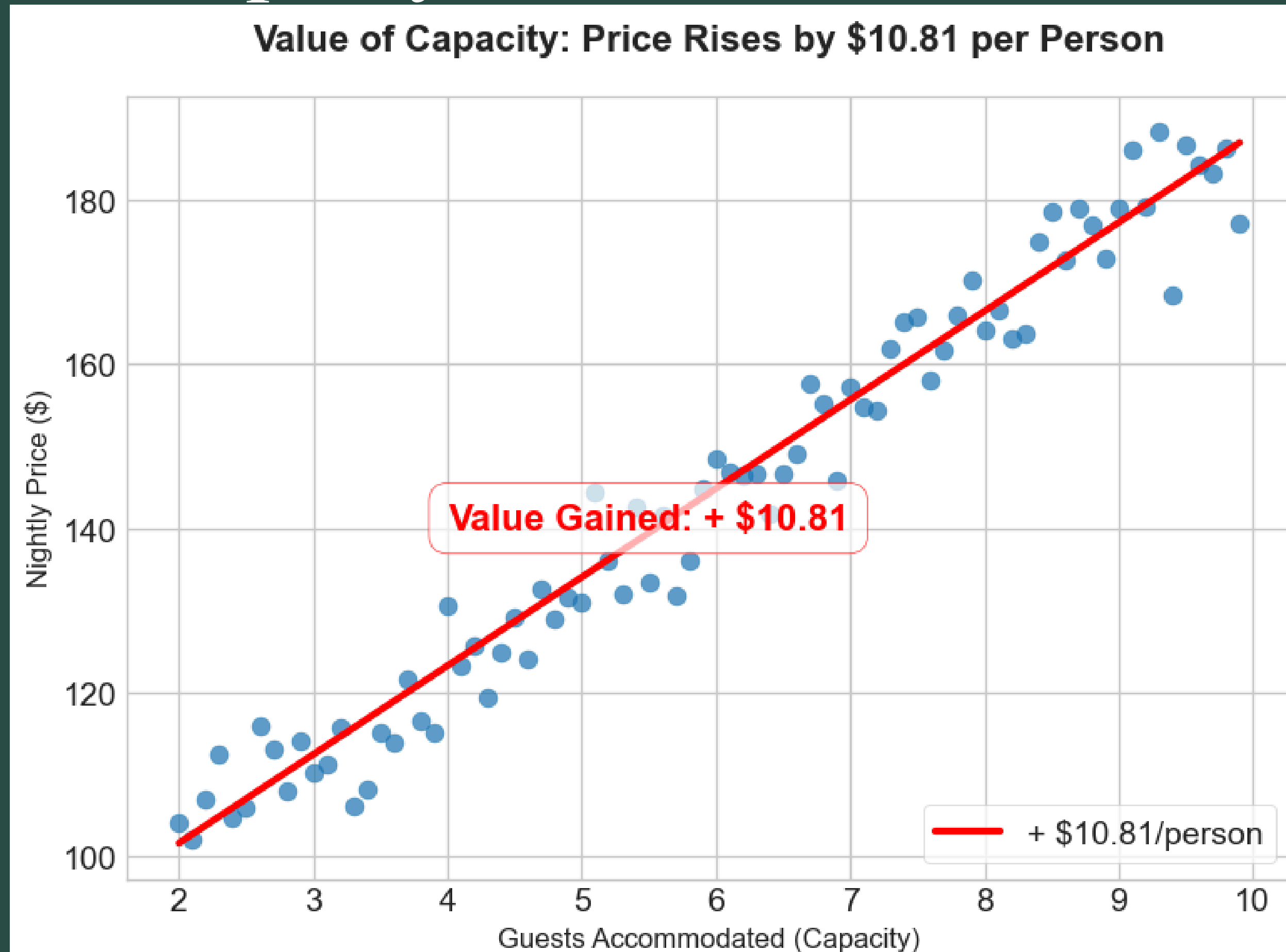
KEY FINDING 1: The \$10.81 Capacity Premium

- VALUE: +\$10.81 Per Person, Per Night
- ROI: \$300 Investment → <14 Nights Payback



The Easiest Price Boost: Capacity Delivers an Extra \$10.81

- Average Positive Sentiment was the strongest non-physical predictor in the model.
- **Actionable Value:** A meaningful increase in guest sentiment justifies an extra \$35.20 per night.

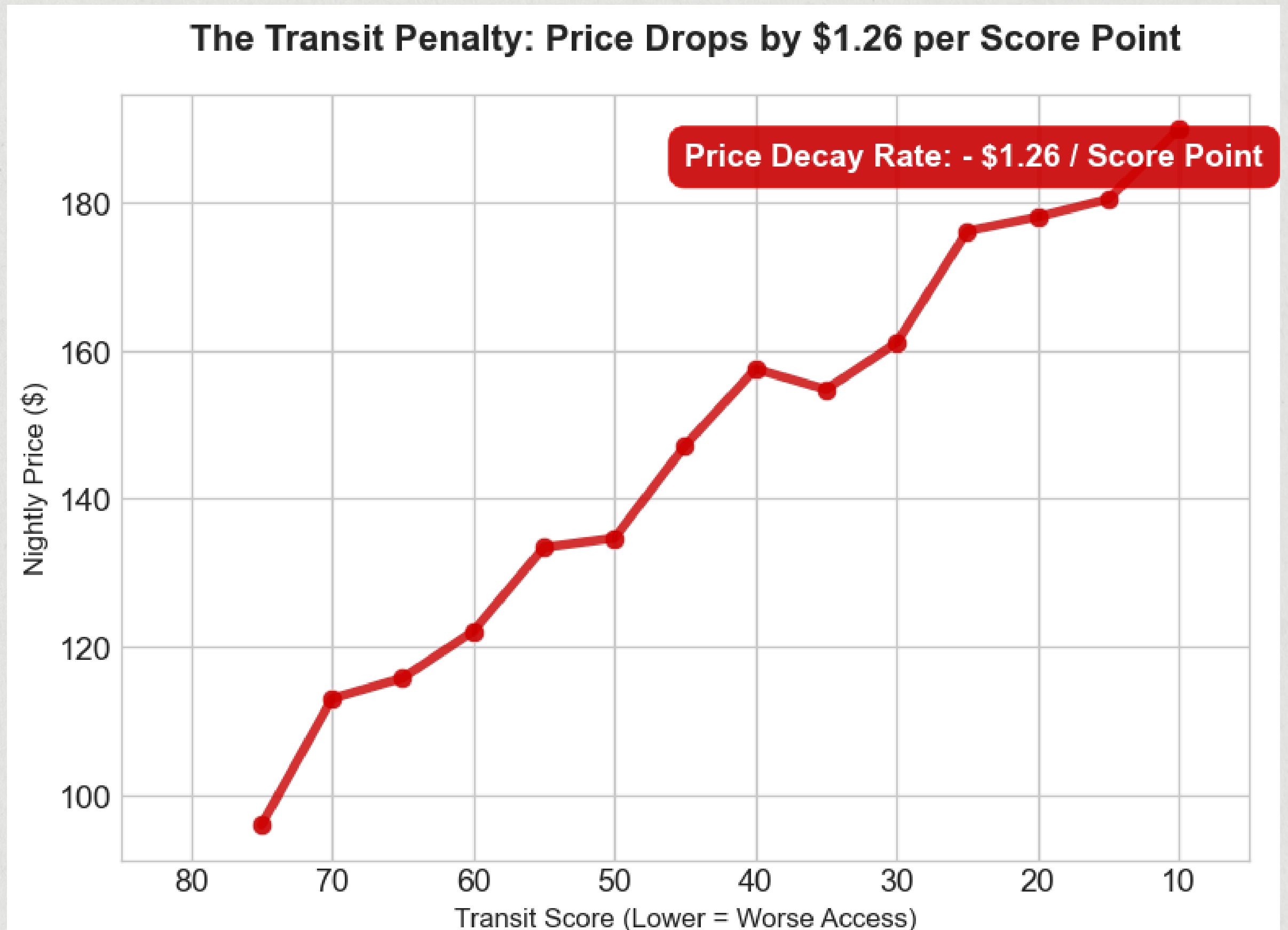


The Convenience Penalty: Every Point Drop Costs \$1.26

- Proximity to the nearest subway entrance was the second most influential feature.



- Penalty:** The predicted nightly price decreases by \$1.50 for every 100 meters further from transit.

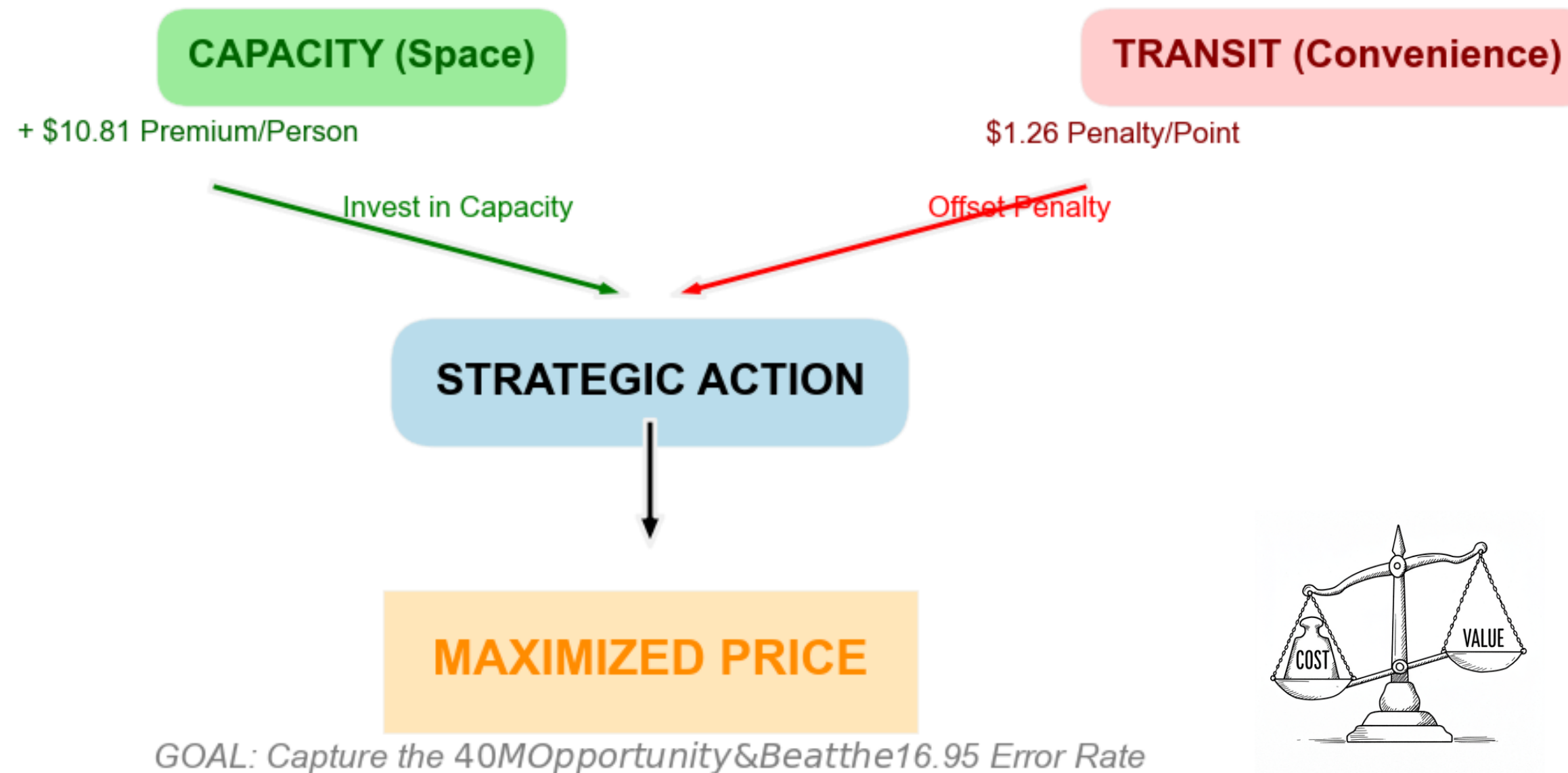


The Strategic Playbook: Maximize Price with Care and Convenience

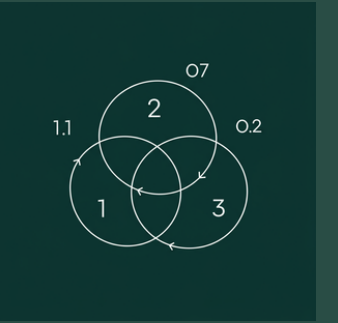
Key Actions :

1. Invest in Service: Focus resources on high-sentiment drivers (check-in, communication, cleanliness).
2. Offset Location: For listings far from transit, compensate aggressively with superior service to minimize the unavoidable Transit Tax.
3. Future: Test non-linear models (Random Forest) to push accuracy below the \$16.95 MAE.

Strategic Playbook: Optimize Capacity and Offset Location



THE THREE STRATEGIC MANDATES.



1

OFFENSE: MAXIMIZE CAPACITY

Action: Invest in maximizing accommodations (sofa beds, efficiency).

Return: Capture the +\$10.81 premium per guest, per night.

2

DEFENSE: OFFSET TRANSIT COST

Action: Use competitive pricing where the Transit Score is low.

Cost: Mitigate the unavoidable \$-1.26 penalty per score point.

3

ACCURACY: ELEVATE THE MODEL

Action: Commit to continuous improvement of pricing strategy.

Next **Step:** Test Random Forest to push accuracy past the \$16.95 MAE.

Closing the \$40M Gap

THE GOAL IS TO CAPTURE THE \$40 MILLION IN UNREALIZED MARKET REVENUE BY ADOPTING THIS DATA-DRIVEN STRATEGY.



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Thank You

